UNIVERSITY of HOUSTON | ECE

ECE 3337: Signals & Systems Analysis

Lecture 13: Laplace Transform

Reference: Chapter 3
(Section 3.6 – 3.8)

Please turn off your gadgets now

Math Moment: Change of Variable

When rewriting a definite integral using a change of variable, it is important to also change the limits.

Example:

$$I = \int_{x=2}^{x=1} 2x \sin(x^2)\, dx$$

Let $u = x^2$, so $du = 2xdx$

$$I = \int_{u=4}^{u=1} \sin(u)\, du$$

Recap: Partial Fractions

The Laplace transform for an LTI system ends up as a ratio of polynomials in s

$$\mathbf{X}(\mathbf{s}) = \frac{\mathbf{N}(\mathbf{s})}{\mathbf{D}(\mathbf{s})} = \frac{b_0 + b_1\mathbf{s} + \ldots + b_m\mathbf{s}^m}{a_0 + a_1\mathbf{s} + \ldots + a_n\mathbf{s}^n}$$

$$= C\,\frac{(\mathbf{s}-z_1)(\mathbf{s}-z_2)\ldots(\mathbf{s}-z_m)}{(\mathbf{s}-p_1)(\mathbf{s}-p_2)\ldots(\mathbf{s}-p_n)}$$

$m < n:$ Strictly proper

$m = n:$ Proper

$m > n:$ Improper

A PARTIAL FRACTION EXPANSION OF THIS RATIO LEADS US TO THE INVERSE TRANSFORM

Big/Messy Partial Fraction Expansions

When expressions get big/messy, it makes sense to use software tools

WolframAlpha computational intelligence

partial fraction expansion (42s^2+78s+150)/(14s(s^2+s(51/14)+50/14))

Extended Keyboard Upload Examples Random

Assuming "s" is a variable | Use "s^2" as a unit instead

Input:

partial fractions	$\dfrac{42\,s^2 + 78\,s + 150}{14\,s\left(s^2 + s \times \frac{51}{14} + \frac{50}{14}\right)}$

Result: Approximate form Step-by-step solution

$$\frac{42\,s^2 + 78\,s + 150}{14\,s\left(s^2 + \frac{51\,s}{14} + \frac{25}{7}\right)} = \frac{3}{s} - \frac{75}{14\,s^2 + 51\,s + 50}$$

Common Inverse Laplace Transforms

$$\mathcal{L}^{-1}\left[\frac{1}{(\mathbf{s}+a)}\right]=e^{-at}u(t)$$

$$\mathcal{L}^{-1}\left[\frac{1}{(\mathbf{s}+a)^2}\right]=te^{-at}u(t)$$

$$\mathcal{L}^{-1}\left[\frac{\omega_0}{(\mathbf{s}+a)^2+\omega_0^2}\right]=e^{-at}\sin(\omega_0 t)u(t)$$

$$\mathcal{L}^{-1}\left[\frac{(\mathbf{s}+a)}{(\mathbf{s}+a)^2+\omega_0^2}\right]=e^{-at}\cos(\omega_0 t)u(t)$$

Convolution in the Laplace Domain

$$y(t) = x(t) * h(t) = \int_{\tau=0-}^{\infty} x(\tau)h(t-\tau)\,d\tau$$

$$\mathcal{L}\left[x(t) * h(t)\right] = \int_{t=0-}^{\infty}\left[\int_{\tau=0-}^{\infty} x(\tau)h(t-\tau)\,d\tau\right]e^{-st}\,dt$$ **Fubini's Theorem**

We can change the order of integration

since t and τ are independent variables

$$= \int_{\tau=0-}^{\infty} x(\tau)\left[\int_{t=0-}^{\infty} \underline{h(t-\tau)e}^{-st}\,dt\right]d\tau$$

This part is just a time shift

Let $\mu = t - \tau$

Convolution in the Laplace Domain

$$\mathcal{L}\left[x(t)*h(t)\right]=\int_{\tau=0-}^{\infty} x(\tau)\left[\int_{\mu=-\tau}^{\infty} h(\mu)e^{-\mathbf{s}(\mu+\tau)}\,d\mu\right]d\tau$$

$$=\int_{\tau=0-}^{\infty} x(\tau)e^{-\mathbf{s}\tau}\,d\tau \int_{\mu=0-}^{\infty} h(\mu)e^{-\mathbf{s}\mu}\,d\mu$$

Due to causality of $h()$

$$=\mathbf{X}(\mathbf{s})\mathbf{H}(\mathbf{s})$$

To convolve two time-domain functions, we can compute their individual Laplace transforms and multiply them and then transform back to the time domain.

The Transfer Function

Input

$\mathbf{X}(\mathbf{s})$

$x(t)$

LTI System

$\mathbf{H}(\mathbf{s})$

Output

$\mathbf{Y}(\mathbf{s}) = \mathbf{H}(\mathbf{s})\mathbf{X}(\mathbf{s})$

$y(t) = h(t) * x(t)$

$$\mathbf{H}(\mathbf{s}) = \frac{\mathbf{Y}(\mathbf{s})}{\mathbf{X}(\mathbf{s})}$$

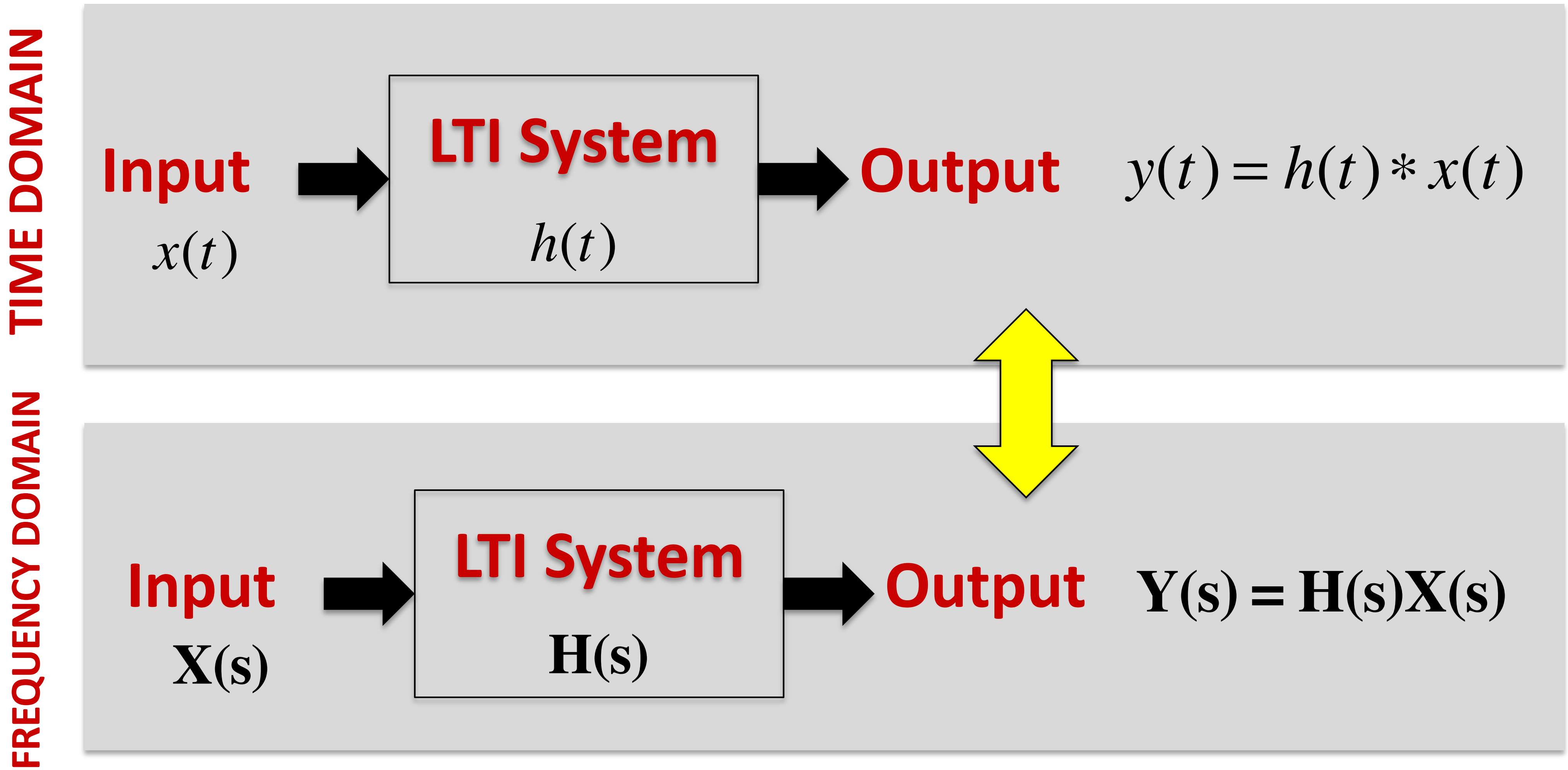
$\mathbf{H}(s)$ is a *<u>complete</u>* description of an LTI system like $\boldsymbol{h}(\boldsymbol{t})$, and has the same information content

Its definition assumes zero initial response, i.e.,

$$y(0^-) = y'(0^-) = \ldots = 0$$

Equivalent System Pictures

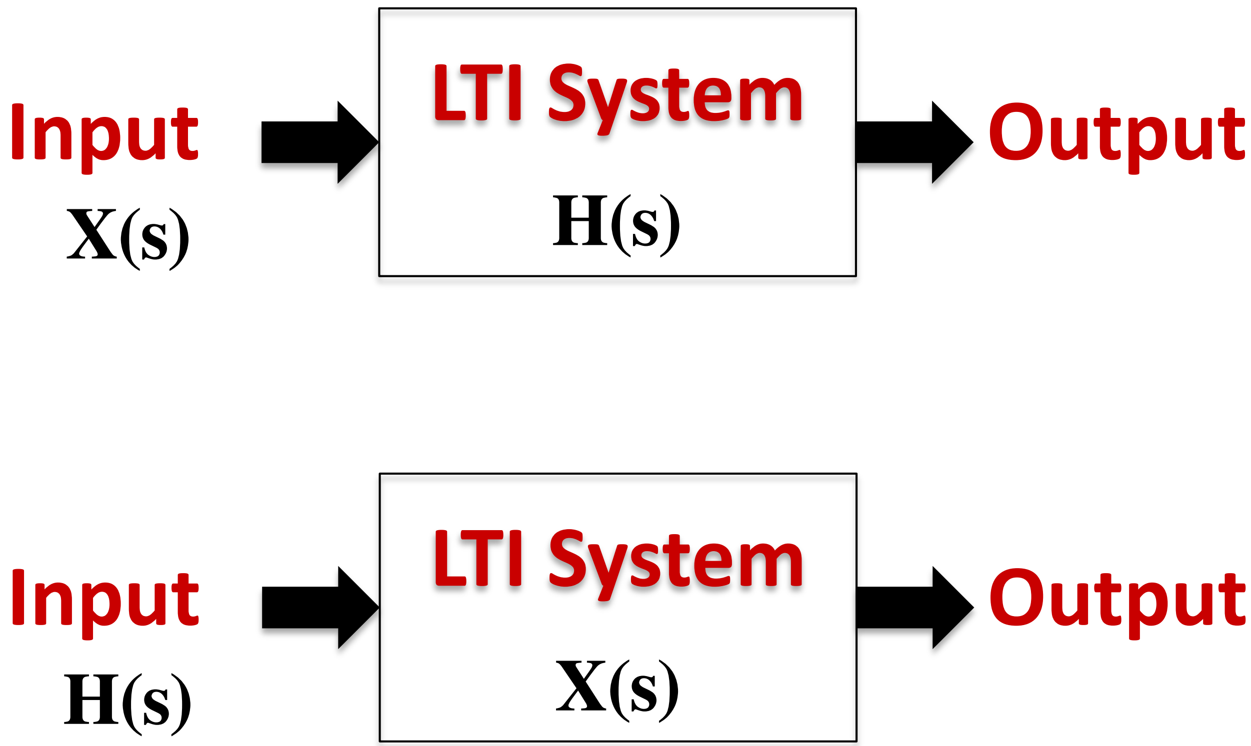
We now have an alternative & equivalent way to describe an LTI system:



System or Signal?

Signals and systems are represented in the same manner, and

- Convolution is commutative, so
- We get the same answer if we switch the signal and the system:



$$Y(s) = H(s)X(s)$$

SAME

$$Y(s) = X(s)H(s)$$
$$= H(s)X(s)$$

Solving LCCDE's using Laplace Transforms

Redo the LCCDE using Laplace Transforms

(assume 0 initial conditions)

$$\frac{d^2y(t)}{dt^2} + a_1\frac{dy(t)}{dt} + a_2y(t) = b_1\frac{dx(t)}{dt} + b_2x(t)$$

Amazing Power !

Let's redo the LCCDE using Laplace Transforms

$$\frac{d^2y(t)}{dt^2}+a_1\frac{dy(t)}{dt}+a_2y(t)=b_1\frac{dx(t)}{dt}+b_2x(t)$$

$$(\mathbf{s}^2+a_1\mathbf{s}+a_2)\mathbf{Y}(\mathbf{s})=(b_1\mathbf{s}+b_2)\mathbf{X}(\mathbf{s})$$

$$\mathbf{H}(\mathbf{s})=\frac{\mathbf{Y}(\mathbf{s})}{\mathbf{X}(\mathbf{s})}=\frac{(b_1\mathbf{s}+b_2)}{(\mathbf{s}^2+a_1\mathbf{s}+a_2)}=\frac{(b_1\mathbf{s}+b_2)}{(\mathbf{s}-p_1)(\mathbf{s}-p_2)}=\frac{A_1}{(\mathbf{s}-p_1)}+\frac{A_2}{(\mathbf{s}-p_2)}$$

$$A_1=(\mathbf{s}-p_1)\mathbf{H}(\mathbf{s})\Big|_{\mathbf{s}=p_1}=\frac{(b_1p_1+b_2)}{(p_1-p_2)}$$

$$A_2=(\mathbf{s}-p_2)\mathbf{H}(\mathbf{s})\Big|_{\mathbf{s}=p_2}=-\frac{(b_1p_2+b_2)}{(p_1-p_2)}$$

$$\Rightarrow h(t)=A_1e^{p_1t}u(t)+A_2e^{p_2t}u(t)$$

WAY EASIER !!

(Adding the initial conditions is easy)

Recap: Table 2-2

Table 2-2: Commonly encountered convolutions.

1. $u(t) * u(t) = t\, u(t)$

2. $e^{at}\, u(t) * u(t) = \left(\dfrac{e^{at}-1}{a}\right) u(t)$

3. $e^{at}\, u(t) * e^{bt}\, u(t) = \left[\dfrac{e^{at}-e^{bt}}{a-b}\right] u(t), \qquad a \neq b$

4. $e^{at}\, u(t) * e^{at}\, u(t) = te^{at}\, u(t)$

5. $te^{at}\, u(t) * e^{bt}\, u(t) = \dfrac{e^{bt}-e^{at}+(a-b)te^{at}}{(a-b)^2}\, u(t),$
   $a \neq b$

6. $te^{at}\, u(t) * e^{at}\, u(t) = \frac{1}{2}\, t^2 e^{at}\, u(t)$

7. $\delta(t-T_1) * \delta(t-T_2) = \delta(t-T_1-T_2)$

LET US TACKLE #5 WITH THE LAPLACE METHOD!

Much Easier

5. $te^{at}\,u(t) * e^{bt}\,u(t) = \dfrac{e^{bt} - e^{at} + (a-b)te^{at}}{(a-b)^2}\,u(t),$

$a \neq b$

$$\mathcal{L}\left[te^{at}u(t)\right] = \frac{1}{(\mathbf{s}-a)^2} \qquad \mathcal{L}\left[e^{bt}u(t)\right] = \frac{1}{\mathbf{s}-b} \qquad \Rightarrow \mathcal{L}\left[y(t)\right] = \frac{1}{(\mathbf{s}-a)^2(\mathbf{s}-b)}$$

$$\frac{1}{(\mathbf{s}-a)^2(\mathbf{s}-b)} = \frac{1}{(a-b)(\mathbf{s}-a)^2} - \frac{1}{(a-b)^2(\mathbf{s}-a)} + \frac{1}{(a-b)^2(\mathbf{s}-b)}$$

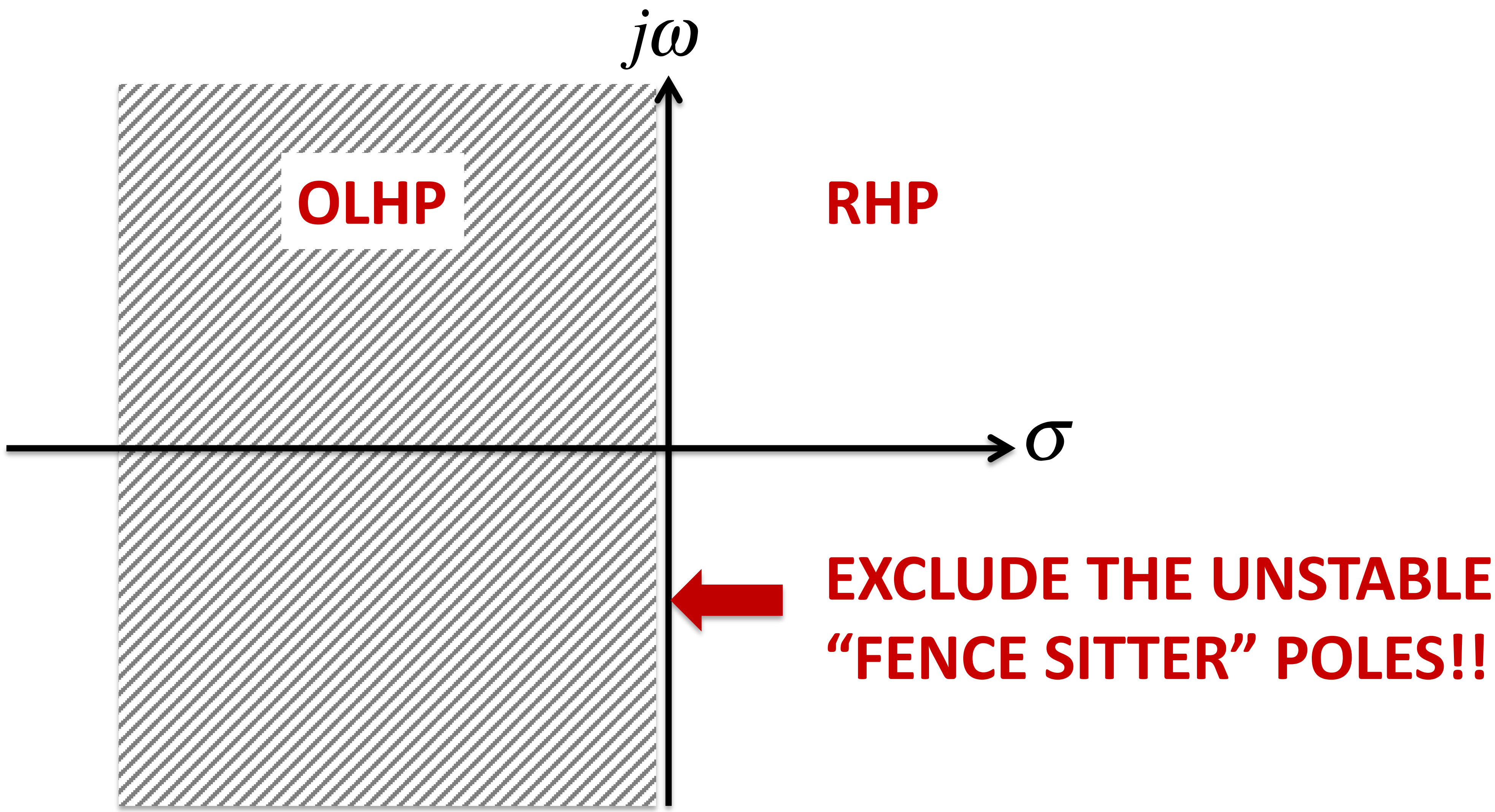
PARTIAL FRACTION EXPANSION

$$\Rightarrow y(t) = \frac{1}{(a-b)}te^{at}u(t) - \frac{1}{(a-b)^2}e^{at}u(t) + \frac{1}{(a-b)^2}e^{bt}u(t)$$

$$= \frac{e^{bt} - e^{at} + (a-b)te^{at}}{(a-b)^2}u(t) \qquad a \neq b$$

Recap: The Open Left Half Plane

The Open Left Half Plane (OLHP) extends from 0- to negative infinity



Interpreting Pole Locations

- A real pole to the left of the vertical axis yields a term with a stable response
- A real pole at the origin yields a term with a ramp response to a step input
- A real pole to the right of the vertical axis is associated with an unstable growing response
- A complex conjugate pair of poles is associated with an oscillatory response
  - When the real part is negative, it is a decaying oscillatory response
  - When the real part is positive, it is a growing oscillatory response
  - When the real part is 0, it is a non-decaying oscillation

Unstable System Example

$$\mathbf{H}(\mathbf{s}) = 1 + \mathbf{s}$$

$$\Rightarrow \mathbf{Y}(\mathbf{s}) = \mathbf{X}(\mathbf{s}) + \mathbf{s}\mathbf{X}(\mathbf{s})$$

$$y(t) = x(t) + \frac{dx(t)}{dt}$$

$$x(t) = x_s(t) + x_n(t)$$

SIGNAL NOISE

$$x_s(t) = 10\sin(1000t)u(t)$$

$$x_n(t) = \frac{1}{100}\sin(10^7 t)u(t)$$

THE NOISE SIGNAL HAS A SMALLER AMPLITUDE, BUT A HIGHER FREQUENCY

Signal to Noise Ratio (SNR)

$$x_s(t) = 10\sin(1000t)u(t)$$

$$x_n(t) = \frac{1}{100}\sin(10^7 t)u(t)$$

POWER:

$$P_s = \frac{1}{2}(10)^2 = 50$$

$$P_n = \frac{1}{2}(10^{-2})^2 = 5\times 10^{-5}$$

POWER RATIO:

$$SNR = \frac{P_s}{P_n} = 10^6$$

REASONABLY CLEAN INPUT

HIGH FREQUENCY NOISE

The Output

$$\mathbf{Y}(\mathbf{s}) = \mathbf{X}(\mathbf{s}) + \mathbf{s}\mathbf{X}(\mathbf{s}) \qquad y(t) = x(t) + \frac{dx(t)}{dt}$$

$$y(t) = x_s(t) + x_n(t) + \frac{d}{dt}\left\{x_s(t) + x_n(t)\right\}$$

$$= 10\sin(10^3 t) + 10^{-2}\sin(10^7 t) + 10^4\cos(10^3 t) + 10^5\cos(10^7 t)$$

$$P_s(@\,\text{output}) = \frac{1}{2}(10^4)^2 + \frac{1}{2}(10)^2 \approx 5\times 10^7$$

$$P_n(@\,\text{output}) = \frac{1}{2}(10^5)^2 + \frac{1}{2}(10^{-2})^2 \approx 5\times 10^9$$

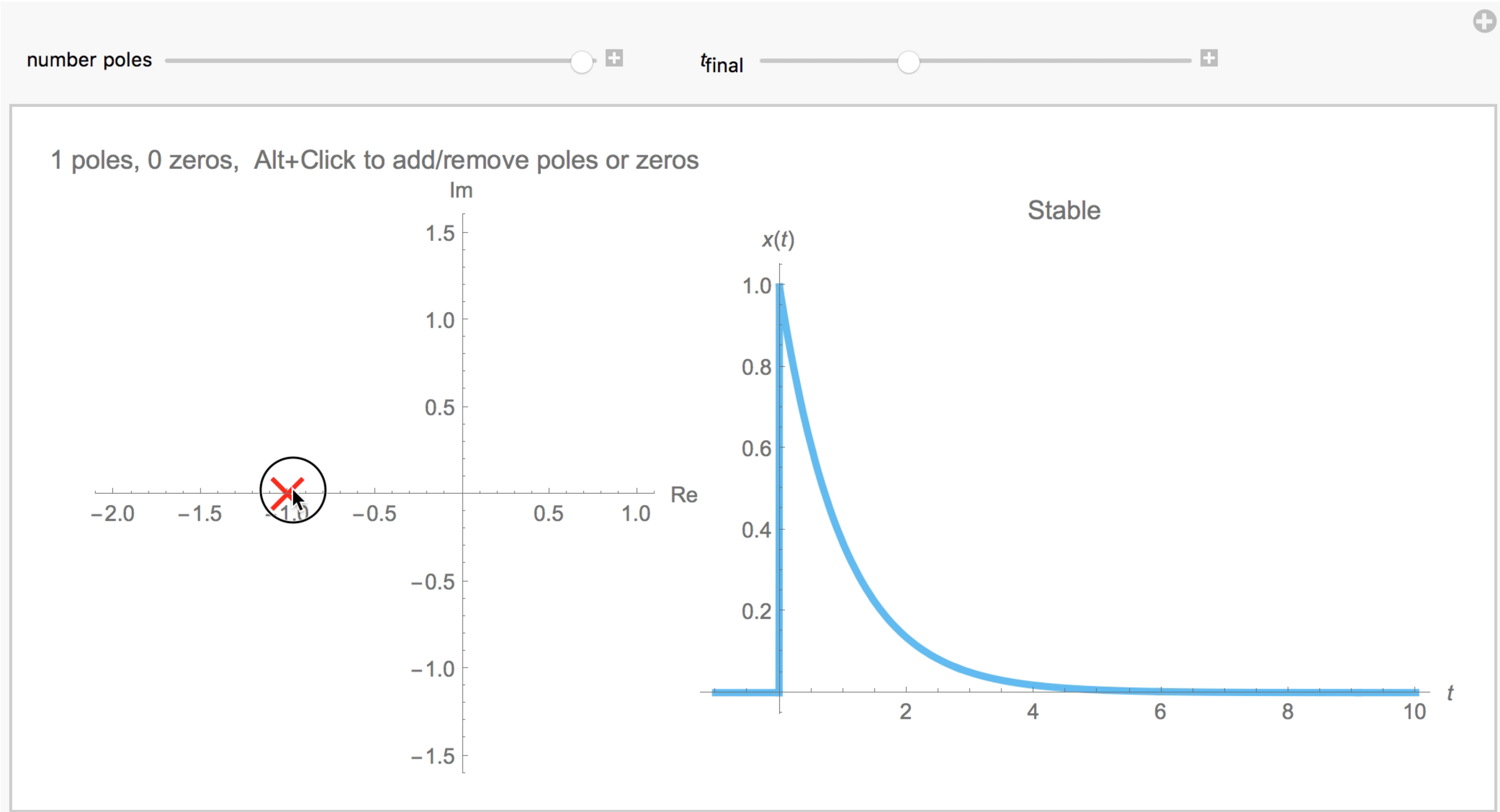
$$SNR = \frac{P_s}{P_n} = 10^{-2}$$

YIKES!!

What Zeros Tell Us

- Zeros do <u>not</u> have any impact on stability, only poles do
- However, zeros affect the coefficients of the partial fraction expansion, and thereby affect the influence of the poles
- We will learn a bit later that zeros on the right half plane are problematic for a reason that is unrelated to stability
  - They produce undesirable non-minimum phase systems that can be confusing to control

Poles and Zeros – The Movie



Repeated Poles

Table 3-3: Transform pairs for four types of poles.

Pole	$\mathbf{X}(\mathbf{s})$	$x(t)$
1. Distinct real	$\dfrac{A}{\mathbf{s}+a}$	$Ae^{-at}\,u(t)$
2. Repeated real	$\dfrac{A}{(\mathbf{s}+a)^n}$	$A\,\dfrac{t^{n-1}}{(n-1)!}\,e^{-at}\,u(t)$
3. Distinct complex	$\left[\dfrac{Ae^{j\theta}}{\mathbf{s}+a+jb}+\dfrac{Ae^{-j\theta}}{\mathbf{s}+a-jb}\right]$	$2Ae^{-at}\cos(bt-\theta)\,u(t)$
4. Repeated complex	$\left[\dfrac{Ae^{j\theta}}{(\mathbf{s}+a+jb)^n}+\dfrac{Ae^{-j\theta}}{(\mathbf{s}+a-jb)^n}\right]$	$\dfrac{2At^{n-1}}{(n-1)!}\,e^{-at}\cos(bt-\theta)\,u(t)$

REPEATED POLES CORRESPOND TO "RAMP LIKE" FACTORS IN THE TIME DOMAIN

System Stability Revisited

We are now ready to take a more general look at system stability

$$\mathbf{X}(\mathbf{s}) = \frac{\mathbf{N}(\mathbf{s})}{\mathbf{D}(\mathbf{s})} = C\frac{(\mathbf{s}-z_1)(\mathbf{s}-z_2)\ldots(\mathbf{s}-z_m)}{(\mathbf{s}-p_1)(\mathbf{s}-p_2)\ldots(\mathbf{s}-p_n)}$$

As with partial fractions, it makes sense to study the answers for strictly proper ($m < n$), proper ($m = n$), and improper functions ($m > n$) separately

Stability Definition

Bounded Input Bounded Output (BIBO)

- If the input is bounded (i.e., finite), then the output will be bounded as well
- This should be true for <u>every</u> bounded input (even a single exception is not possible)
- LTI systems for which this is true are said to be BIBO Stable

For a BIBO system, the impulse response should be absolutely integrable:

$$\int_{-\infty}^{\infty} |h(t)|\,dt \text{ is finite}$$

Practical Stability Testing

Test Input $u(t)$ → LTI System $h(t)$ → Output $y_{STEP}(t) = h(t) * u(t)$

The unit step function is a commonly preferred input for stability testing

- It is bounded (max = 1)
- It has infinite energy
- It is mathematically simple
- It is easy to generate in the laboratory (flip a switch ON)

Strictly Proper Case (m < n)

Again, it is best to learn from examples

$$\mathbf{H}(\mathbf{s}) = \frac{\mathbf{s}^2 - 4\mathbf{s} + 3}{\mathbf{s}(\mathbf{s}+1)(\mathbf{s}+3)} = \frac{1}{\mathbf{s}} - \frac{4}{\mathbf{s}+1} + \frac{4}{\mathbf{s}+3}$$

$$\Rightarrow h(t) = [1 - 4e^{-t} + 4e^{-3t}]u(t)$$

- **Poles at 0, -1, and -3**
- **Let's apply our favorite bounded input, the unit step function, and see what happens:**

$$y_{STEP}(t) = h(t) * u(t)$$

Strictly Proper Case (m < n)

$$\mathcal{L}[u(t)] = \frac{1}{s}$$

$$\mathbf{H}(\mathbf{s}) = \frac{\mathbf{s}^2 - 4\mathbf{s} + 3}{\mathbf{s}(\mathbf{s}+1)(\mathbf{s}+3)} = \frac{1}{\mathbf{s}} - \frac{4}{\mathbf{s}+1} + \frac{4}{\mathbf{s}+3}$$

$$\mathbf{Y}(s) = \underline{\frac{1}{s^2}} - \frac{4}{s(s+1)} + \frac{4}{s(s+3)}$$

Even before completing the partial fraction expansions, we see right away that the first term is problematic, since it corresponds to a ramp that grows indefinitely !!

In the Time Domain: Table 2-2

Table 2-2: Commonly encountered convolutions.

1. $u(t) * u(t) = t\, u(t)$

2. $e^{at}\, u(t) * u(t) = \left(\dfrac{e^{at} - 1}{a} \right) u(t)$

3. $e^{at}\, u(t) * e^{bt}\, u(t) = \left[\dfrac{e^{at} - e^{bt}}{a - b} \right] u(t), \qquad a \neq b$

4. $e^{at}\, u(t) * e^{at}\, u(t) = t e^{at}\, u(t)$

5. $t e^{at}\, u(t) * e^{bt}\, u(t) = \dfrac{e^{bt} - e^{at} + (a-b) t e^{at}}{(a-b)^2}\, u(t),$
$a \neq b$

6. $t e^{at}\, u(t) * e^{at}\, u(t) = \frac{1}{2} t^2 e^{at}\, u(t)$

7. $\delta(t - T_1) * \delta(t - T_2) = \delta(t - T_1 - T_2)$

Strictly Proper Case (m < n)

We put Table 2-2 to use

$$y_{STEP}(t) = [1 - 4e^{-t} + 4e^{-3t}]u(t) * u(t)$$

$$= u(t) * u(t) - 4[e^{-t}u(t)] * u(t) + 4[e^{-3t}u(t)] * u(t)$$

$$= \underline{tu(t)} - 4\frac{(e^{-t} - 1)}{(-1)}u(t) + 4\frac{(e^{-3t} - 1)}{(-3)}u(t)$$

- The underlined ramp term grows indefinitely as *t* increases, so **NOT BIBO STABLE !**
- **The problematic term was the pole at 0**

Proper Case (*m = n*)

Example:

$$\mathbf{H}(\mathbf{s}) = \frac{2\mathbf{s}^2 + 10\mathbf{s} + 6}{\mathbf{s}^2 + 5\mathbf{s} + 6} = 2 - \frac{6}{\mathbf{s}+2} + \frac{6}{\mathbf{s}+3}$$

- We already know that the second and third terms lead to (unproblematic) damped responses, so let's examine the constant term.
- The step response is just a finite constant (safe)

Basically, the proper case is no different from the strictly proper case, i.e., poles need to be in the Open LHP for BIBO stability

Improper Case ($m > n$)

Example:

$$\mathbf{H}(\mathbf{s}) = \frac{6\mathbf{s}^3 + 4\mathbf{s}^2 + 8\mathbf{s} + 6}{\mathbf{s}^2 + 2\mathbf{s} + 1} = \underline{6\mathbf{s}} - 8 + \frac{18\mathbf{s} + 14}{(\mathbf{s}+1)^2}$$

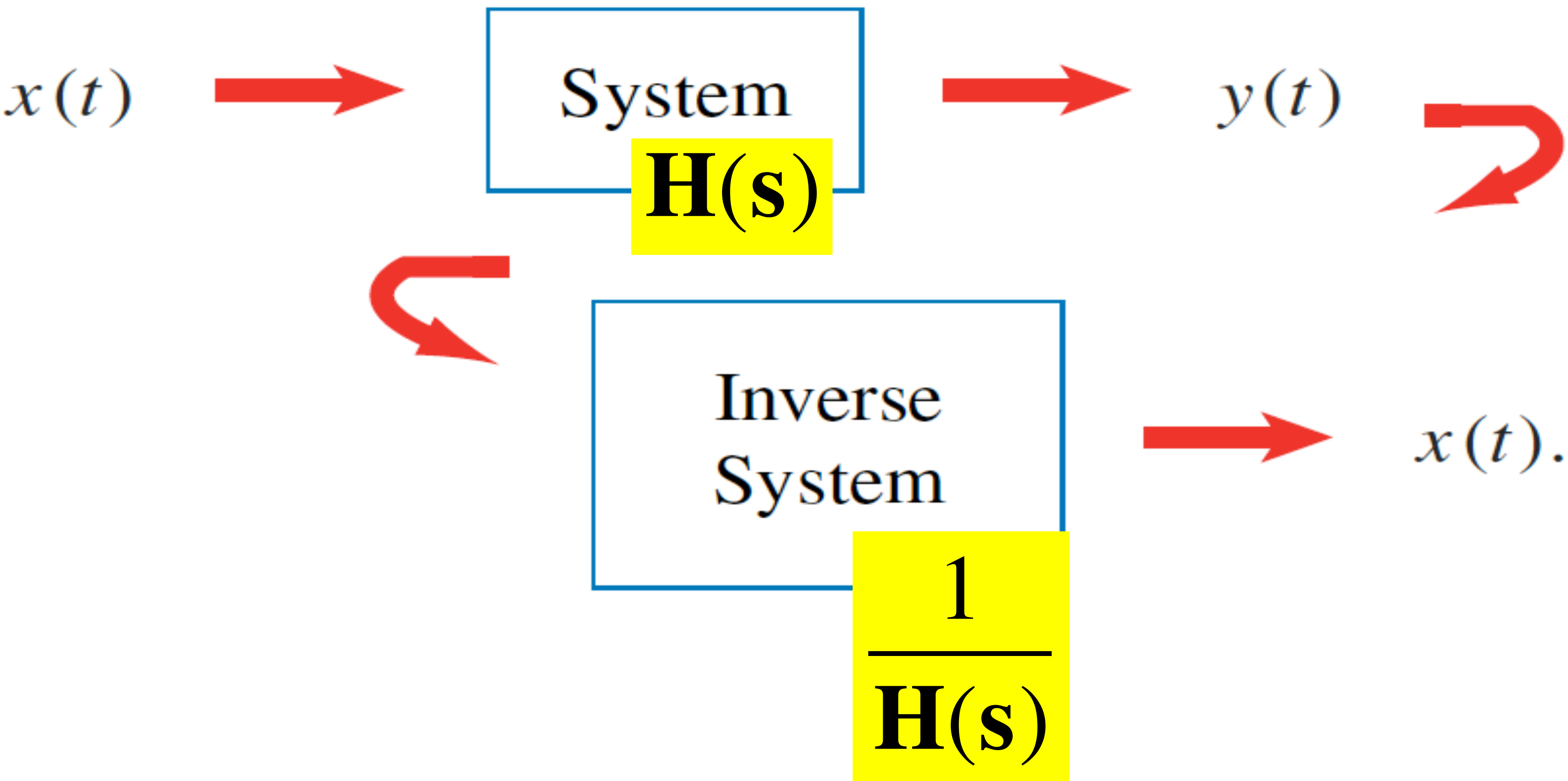
The potentially problematic term is the first one (6s), so let's see what happens when we apply a step input (1/s). The output is:

$$y(t) = \mathcal{L}^{-1}\left[6\mathbf{s} \bullet \frac{1}{\mathbf{s}}\right] = 6\,\delta(t) \quad \textbf{INFINITE}$$

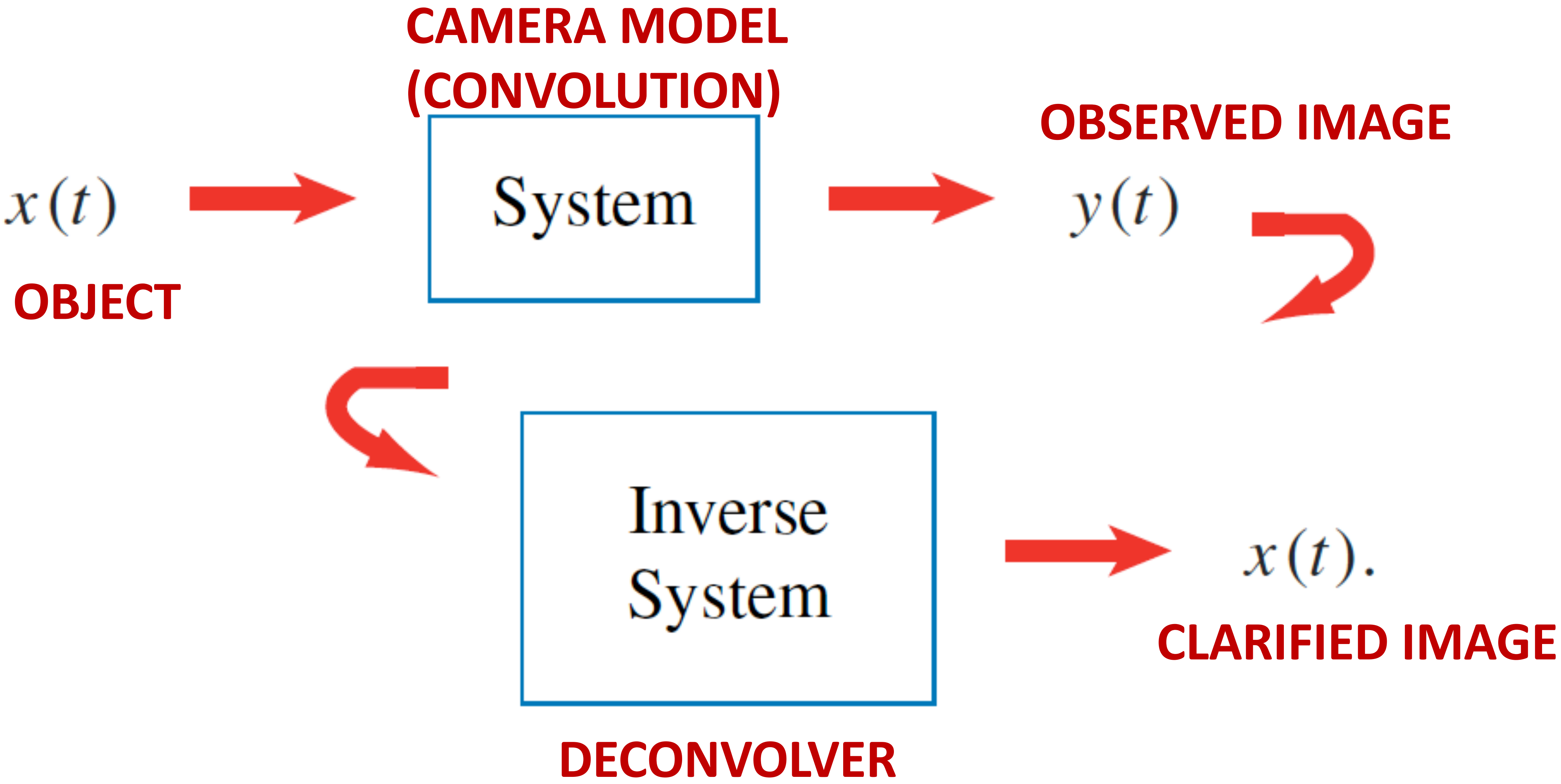
Improper transfer functions are <u>always</u> BIBO unstable, regardless of the locations of their poles and zeros

Invertible Systems

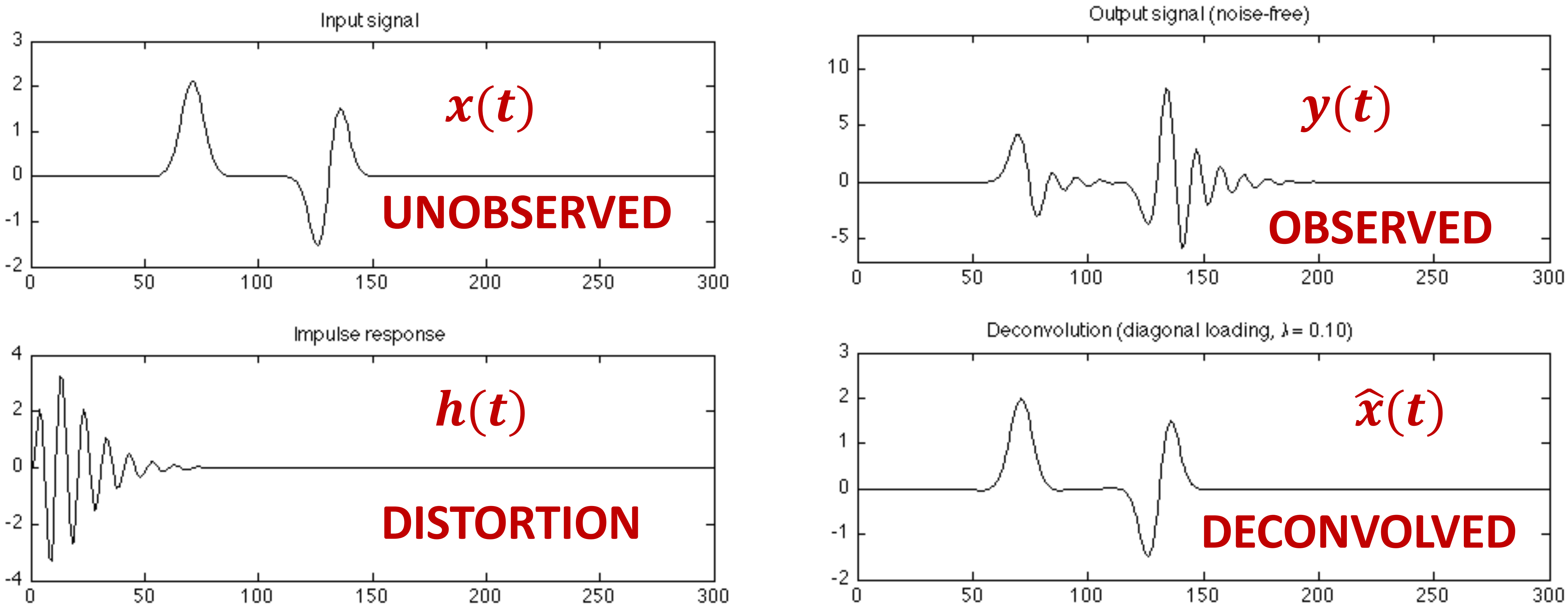
Mathematically, the inverse system is easy to write down:



Deconvolution

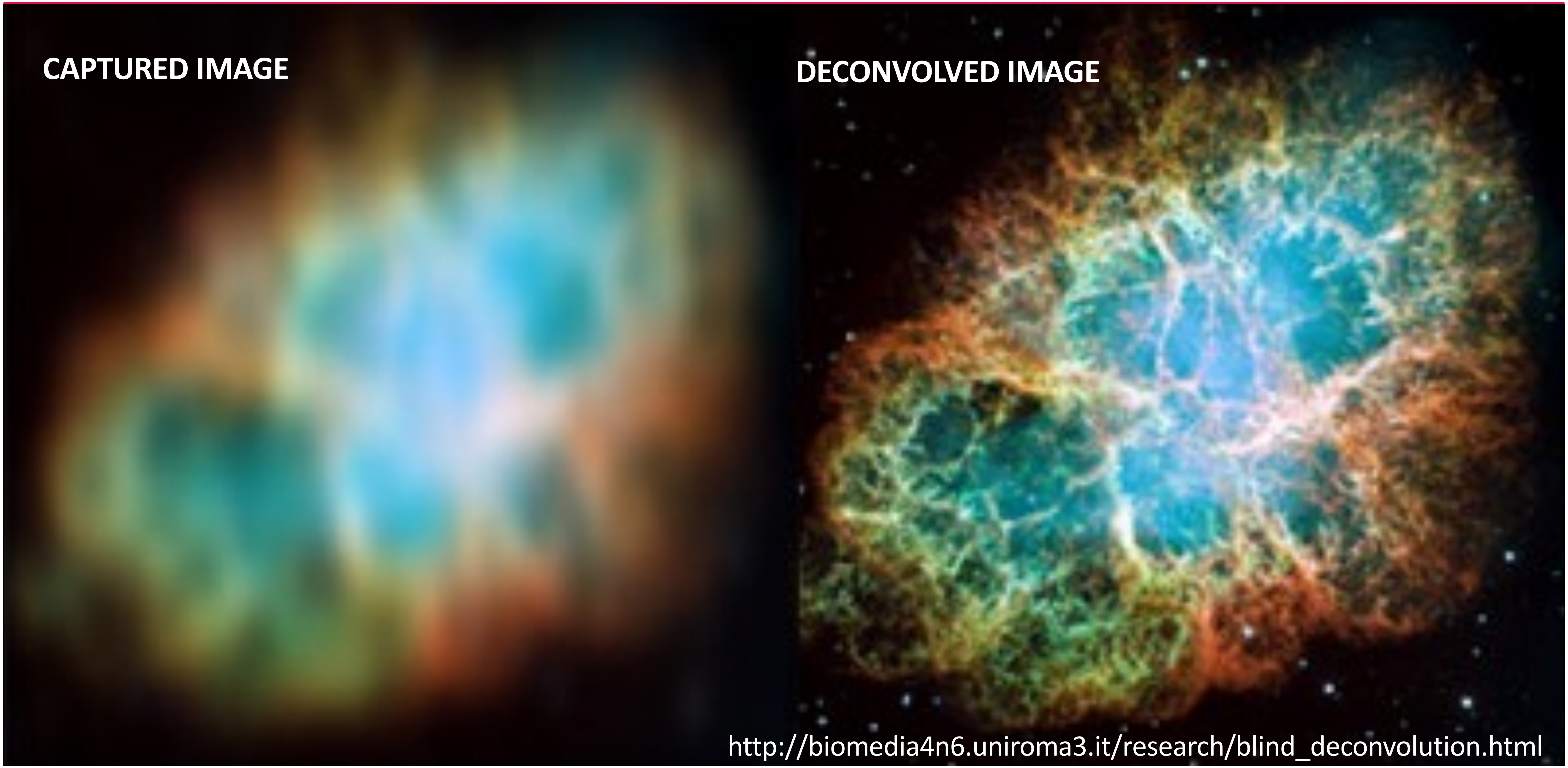


1-D Signal Example

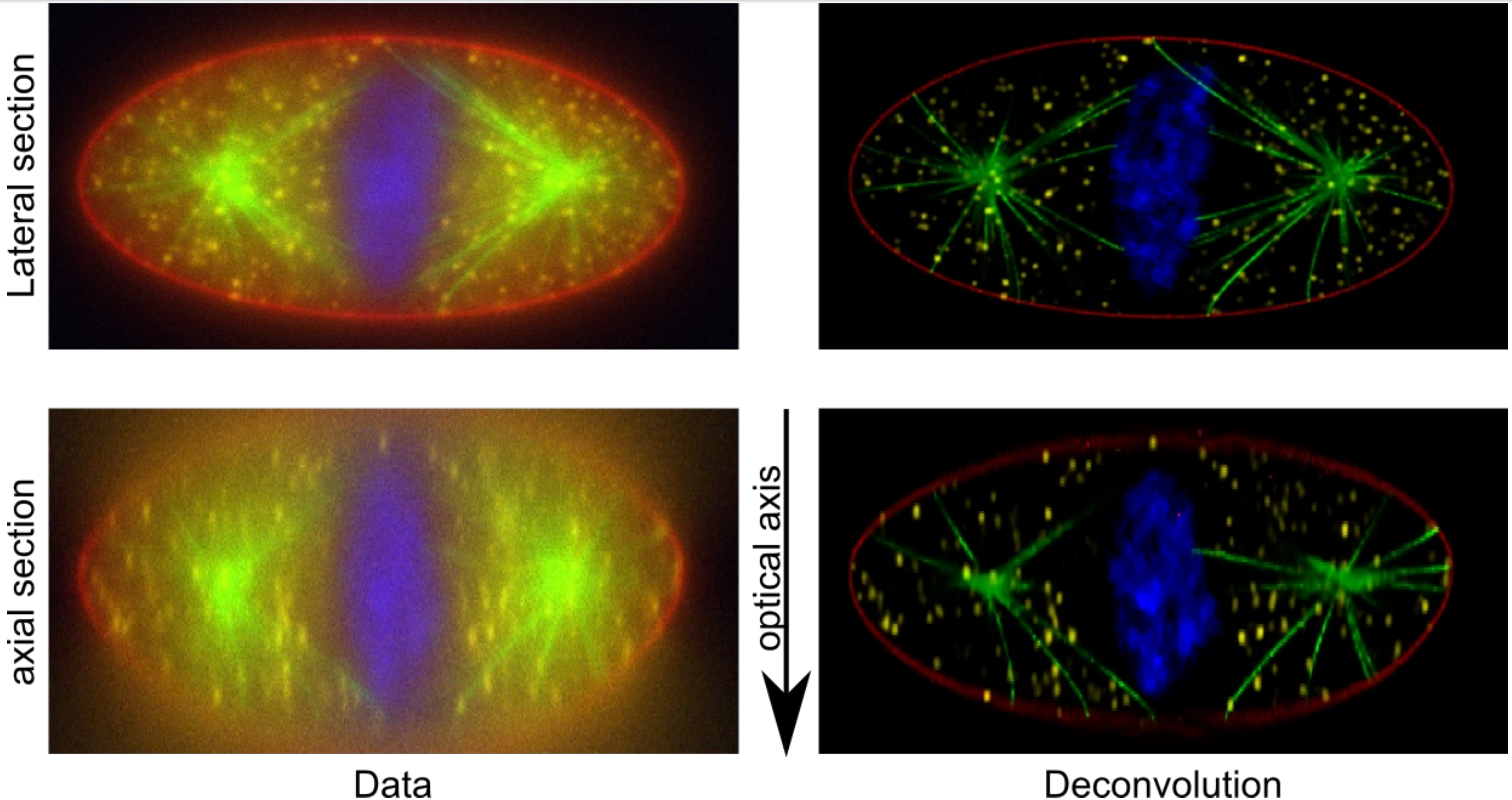


https://eeweb.engineering.nyu.edu/iselesni/lecture_notes/least_squares/LeastSquares_SPdemos/deconvolution/html/deconv_demo.html

Example: Deconvolution in Astronomy



Deconvolution in Microscopy



https://cral-perso.univ-lyon1.fr/labo/perso/ferreol.soulez/coupesResultat_en.png

So, what's the catch?

The inverse system must be well behaved

$$\frac{1}{H(s)}$$

The poles and zeros of the System and the Inverse system are swapped, so the Inverse System may or may not be well behaved

- It may not be stable
- It may not be causal

Opportunity

$$\boxed{\frac{1}{H(s)}}$$

Interestingly, the majority of deconvolution applications have challenging transfer functions, and added noise

- We also have to contend with noise and signal distortions that may be non-linear
- A rich area for signal processing research: Opportunity to invent custom deconvolution algorithms for applications

Theorem

▶ A BIBO stable and causal LTI system has a BIBO stable and causal inverse system if and only if all of its poles and zeros are in the open left half-plane, and they are equal in number (its transfer function is proper). Such a system is called a ***minimum phase*** system. ◀

MINIMUM PHASE SYSTEMS BECOME IMPORTANT WHEN DESIGNING CONTROL SYSTEMS.

NON-MINIMUM PHASE SYSTEMS ARE TRICKY TO CONTROL

Example

DOES THE FOLLOWING SYSTEM MEET THE CRITERIA FOR INVERTIBILITY?
IF SO, DERIVE THE INVERSE SYSTEM IN THE SAME FORMAT

$$h(t) = \delta(t) + 2e^{-3t}u(t) - 6e^{-4t}u(t)$$

STEP 1: Write down the transfer function:

$$\mathbf{H}(\mathbf{s}) = 1 + \frac{2}{\mathbf{s}+3} - \frac{6}{\mathbf{s}+4}$$

$$= \frac{(\mathbf{s}+1)(\mathbf{s}+2)}{(\mathbf{s}+3)(\mathbf{s}+4)}$$

Example (contd.)

STEP 2: Analyze the zeros and poles

- ✓ **H**(**s**) is is proper ($m = n = 2$)
- ✓ Zeros at {-1, -2}, both in the open left-half plane
- ✓ Poles at {-3, -4}, both in the open left-half plane

Therefore, the inverse exists, and H(s) is a minimum phase system

STEP 4: Derive the inverse system

$$\mathbf{G}(\mathbf{s}) = \frac{1}{\mathbf{H}(\mathbf{s})} = \frac{(\mathbf{s}+3)(\mathbf{s}+4)}{(\mathbf{s}+1)(\mathbf{s}+2)}$$

$$g(t) = \delta(t) + 6e^{-t}u(t) - 2e^{-2t}u(t)$$

Non-minimum Phase Systems

Let's tweak the system and add a zero to the right half plane (the poles remain the same)

$$\mathbf{H}_{NEW}(\mathbf{s}) = \frac{(\mathbf{s}+1)(\mathbf{s}+2)(1-\mathbf{s})}{(\mathbf{s}+3)(\mathbf{s}+4)}$$

$$= \mathbf{H}(\mathbf{s}) - \mathbf{sH}(\mathbf{s})$$

$$h_{NEW}(t) = h(t) - \frac{dh(t)}{dt}$$

ASSUME ZERO INITIAL CONDITION

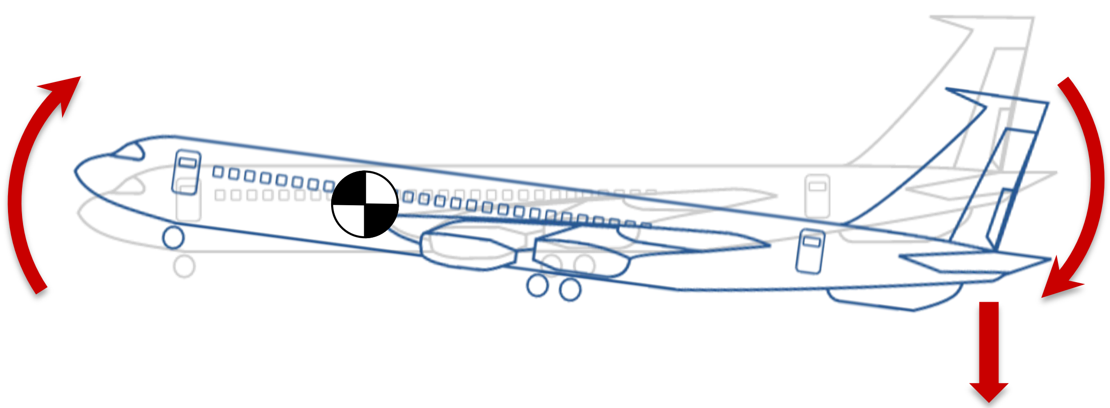
NON-MINIMUM PHASE SYSTEMS ARE CONFUSING TO CONTROL

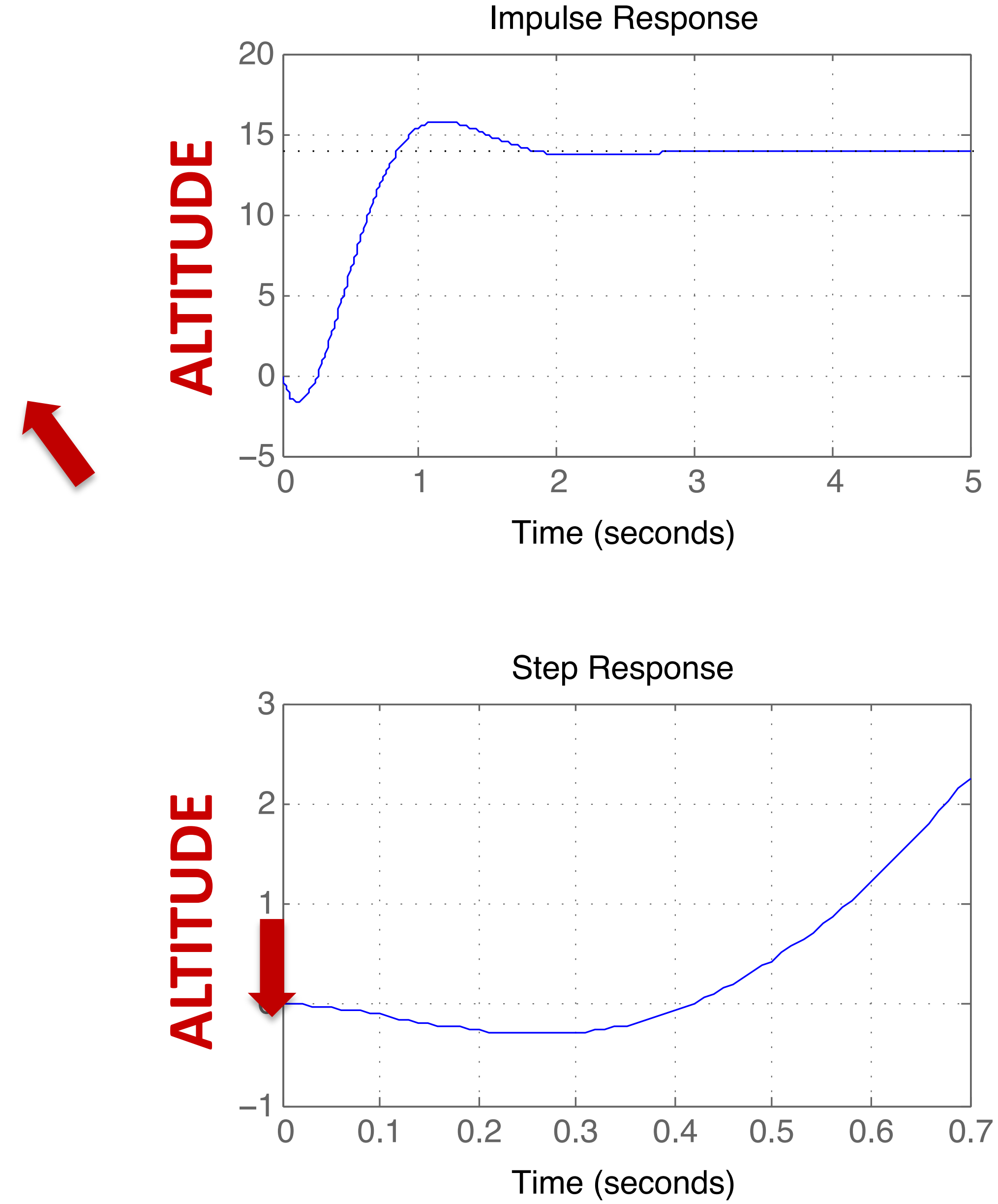
Airplane Control

$$H(s) = \frac{Y(s)}{E(s)} = \frac{30(s-6)}{s(s^2+4s+13)}$$

$Y(s)$: Altitude

$E(s)$: Elevon Deflection





Airplane Canards

Some fighter airplanes have canards to suppress non-minimum phase behavior

Other Real-life Examples

- **Bicycles:** To turn a bike left, you need to lean slightly right initially
- **Parallel parking a car close to the curb:** The transfer function from steering angle to distance from the curb is non-minimum phase.
- **Airplanes:** The transfer function from elevon to altitude is non-minimum phase. When the elevon is raised there is a force that pushes the rear of the airplane down. This causes a rotation which gives an increase of the angle of attack and an increase of the lift. Initially the aircraft will lose altitude (confusing). *The Wright brothers understood this weird effect and added control surfaces in the front of the aircraft to suppress it*.
- **Balancing a stick on your finger and trying to move it:** You need to move it backwards initially
- **Hoverboards and Segways:** To move forward, you may need to lean backwards initially

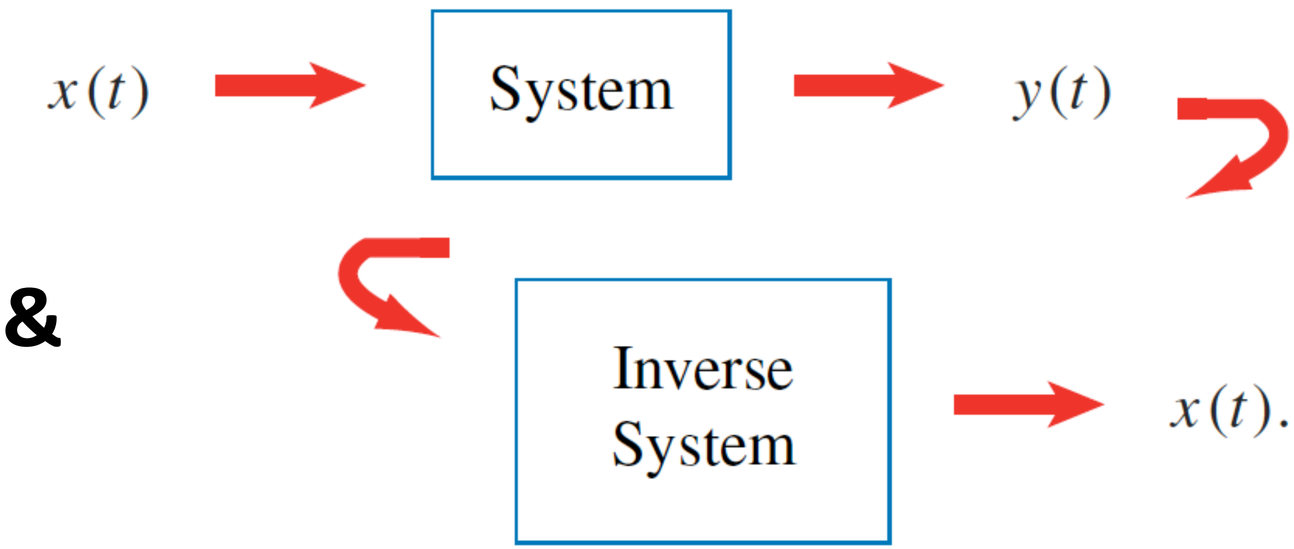
Quick Take

- The presence of complex conjugate pole pairs indicates an oscillatory nature of the system
  - Left of the vertical axis $\Rightarrow$ dying oscillation
  - On the vertical axis $\Rightarrow$ stable oscillation
  - Right of the vertical axis $\Rightarrow$ growing oscillation
- Improper transfer functions are always BIBO unstable, no further analysis is needed!
- Proper and strictly proper transfer functions are BIBO stable if all the poles are in the open left-half plane
- Proper systems are minimum phase if all poles and zeros are in the open Left half plane, and equal in number

Summary

The structure of the transfer function tells us a lot about its stability

- **Locations and poles and zeros**
- **We looked at inverse systems & minimum phase systems**
- **The structure of the Laplace transform for the inverse system tells us a lot about its properties**



For the Next Class

- Work through the textbook examples in Sections 3.6 – 3.8
- Study Sections 3.10-3.11 (we are skipping Section 3-9 of the textbook)

How to Study:

1. Read the assigned sections in order to understand the basic concepts, and aim to memorize the concepts
2. Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test
3. Write down your questions and bring them to class